

SUPER CODER 3.0



Name _____

Class _____ Section _____

Roll No. _____ Phone No _____

School _____



SuperCoder 3.0 | First Edition 2022 | Revised Edition 2025

As per latest guidelines given in National Skills Qualifications Framework
and the Revised Skill Education Curriculum

Aligned with NEP 2020, NCF 2023

PREFACE

In the rapidly evolving digital landscape of the 21st century, technological proficiency has become a cornerstone of modern education and professional development. As our world becomes increasingly interconnected and digitally driven, the ability to understand, navigate, and create within the digital realm is no longer a luxury, but a necessity. This shift demands a comprehensive approach to both digital literacy and coding education.

SuperCoder 3.0, developed by SuperTeacher Edureforms Pvt Ltd, offers such a curriculum designed to meet this growing need for digital competencies. This course is founded on robust educational principles, based on international standards (CSTA and ISTE), and aligned with the National Skills Qualifications Framework, the Revised Skill Education Curriculum, India's National Education Policy 2020, and the National Curriculum Framework 2023. It employs a progressive learning approach, with each level building upon the previous one.

At its core lies the **C4 model** - Context, Connection, Concept, and Competency - the load-bearing pillars that support not just the retention of information, but its practical application in the real world. This is not mere rote learning, but a living, breathing understanding of technology's place in our lives.

SuperCoder 3.0 encompasses a broad spectrum of digital knowledge areas: Digital Literacy for foundational skills, Programming and Computational Thinking for logical problem-solving, Artificial Intelligence for understanding intelligent systems, Data Science for extracting insights, Digital Citizenship and Cybersecurity for safe navigation, and Gaming and Game Development as a powerful dimension that harnesses interactive entertainment to teach design thinking, creative problem-solving, and advanced programming concepts through engaging, hands-on experiences.

It also includes skill modules recommended by experts, ensuring that the approach produces well-rounded digital citizens equipped to navigate the complexities of our society. The skills and knowledge imparted through SuperCoder 3.0 serve as the foundational elements for future innovations. They provide learners with the tools to conceptualize, design, and implement technological solutions to complex problems.

With **SuperCoder 3.0** as your blueprint, what wonders will you build? What problems will you solve? What future will you create? We invite you on this educational journey!

The SuperCoder Team
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OUR TEAM

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CONTENT FEATURES

Hi! I'm AI-Ra



I am a new creation, inspired by the A.I (Artificial Intelligence) and Ra, the Egyptian god of sun and light, both of which represent 'knowledge'. I'll be with you throughout the book, explaining the concepts in a flowing, lucid way. This textbook is written using the C4 Model: Context Setting, Connecting to Real Life, Concept Exploration, and Competency Building.



The chapters begin with context-setting: a question which is answered at the end. It can be a problem statement, or a topic-related intriguing question to spark the readers' curiosity and learning goals of the chapter in brief.

InfoByte



The InfoBytes contain interesting facts related to the topic to hook the learners to the lesson.

try < it >



There will be many 'Try it's' sprinkled in between the chapter body. They are simple activities to keep the students' minds stimulated.



This definition boxes are used whenever a new term is introduced. These define the term, without breaking the flow of the lesson.

LET'S RECAP!

'Let's Recap' summarizes the important points taught in the chapter.

WORKSHEET



The worksheet has varied types of questions, to assess the learning. The questions check the competencies of the learners, rather than their rote-learning ability.

ACTIVITY JUNCTION!

'Activity Junction' directs the students to explore and improve their skills, in context of the topics taught in the chapter.

This book series integrates NEP, SDGs, Competency-Based Education frameworks, CSTA and ISTE standards throughout its content, aligning with modern educational approaches and global standards while fostering comprehensive skill development.

DOMAIN I : DIGITAL LITERACY

1. CREATE WITH KRITA - Simple Sketch using Krita 5

- Digital painting
- Getting started with Krita
- Types of layers
- Toolbox
- Brushes, Presets and Stamps
- Project: Starry Night

2. THE LANGUAGE OF THE UNIVERSE - Understanding the Number Systems 26

- Why numbers?
- Numeral system
- Decimal system
- Binary system
- Octal system
- Hexadecimal system

DOMAIN II : PROGRAMMING AND COMPUTATIONAL THINKING

3. PLAY WITH PYTHON - Basics of Python 43

- Getting Started with Python
- Print() Function
- Interactive Mode and Script Mode
- Project 1: Student ID Card
- Variables and Constants
- Data Types and Operators
- Project 2: Result Generator

4. IF YOU THINK FOR A WHILE - Loops and Conditionals in Python 67

- Conditional statements
- If and If-else statements
- Loops
- For and While loops
- Project 1: Multiplication table
- Project 2: Guess who am I?

5. TURTLE WITH PYTHON - Introduction to Functions and Libraries in Python 87

- Functions
- Project 1: About Me
- Modules
- Project 2: Square and Square Root of a Number
- Project 3: Guess the Number!
- Libraries
- Turtle Graphics Library

6. MIGHTY APPS WITH MIT APP INVENTOR - Creating Simple Apps 111

- Components and types of apps
- Creation of an app
- MIT App Inventor
- Exploring MIT App Inventor
- Project 1: Grow and Shrink
- Project 2: Flowers

7. MAKE A WIKI- Understanding HTML 133

- Web Pages and Websites
- HTML Tags
- HTML Attributes
- Project: MS Dhoni

DOMAIN III : ARTIFICIAL INTELLIGENCE

8. AI FOR A BETTER WORLD - Understanding AI Domains and Ethics 162

- Artificial Intelligence (AI)
- Domains of AI
- AI for SDGs
- AI Ethics
- System Maps

DOMAIN IV : DATA SCIENCE

9. SMALL STEPS TO BIG DATA - Introduction to the Big Data 182

- Using data in real life
- Data science
- Types of data & data collection
- Big Data and its characteristics (5V)
- Applications of Big Data
- Pivot tables
- Project: Class Fruit Survey with a Pivot Table

DOMAIN V : DIGITAL CITIZENSHIP AND CYBERSECURITY

10. CREATING A POSITIVE DIGITAL FOOTPRINT - Responsible Online Behaviour 198

- Your online identity
- Going anonymous
- Digital footprint
- Advantages of positive digital footprint